How Chatbots Work KEY

Name _____

Work with a partner. Compare answers when you finish each problem.

Word bank: neural network next word tokens true word

1. Finish the sentence. One word each.

Peanut butter and jelly Once upon a time I am going to the Answers vary.

2. Count what comes next. The training text:

> I like rice. I like soccer. I like rice and beans. My friend likes soccer. My friend likes rice.

Finish the table. Count how many times each word comes next. ("end" means the end of a sentence.)

After this word	These words came next (how many times)	Dice
I	like (3)	always "like"
like	rice (2), soccer (1)	1 to 4: rice, 5 or 6: soccer
likes	soccer (1), rice (1)	1 to 3: soccer, 4 to 6: rice
rice	end (2), and (1)	1 to 4: end, 5 or 6: and
soccer	end (2)	always end
and	beans (1)	always "beans"
beans	end (1)	always end
My	friend (2)	always "friend"
friend	likes (2)	always "likes"

The dice column gives more common words more numbers. After "like": rice 67 %, soccer 33 %

3. Tiny chatbot, real chatbot.

Both chatbots predict the next word. Neither one checks if a sentence is true.

A real chatbot reads tokens: words and pieces of words.

A real chatbot is a neural network. Its output layer has one neuron for every word.

4. The dice chatbot. Start with **I** or **My**. Roll a die when there's a choice.

Sentence 1: Answers vary. Example: I like rice and beans.

Sentence 2: _____

Sentence 3: _____

Did you write a sentence that is **not** in the training text? Which one? My friend likes rice and beans.

5. Talk about it. Did the tiny chatbot understand "My friend likes rice and beans"?

The tiny chatbot **did** / **did not** understand, because Answers vary. It only counted.

_____

_____

Go Further (Day 2). The word-counting chatbot on screen.

With "no chance," is the sentence the same every time? yes

With "a lot of chance," the bars get flatter (more even) . Rare words win more often .

A strange sentence it wrote: Answers vary.

ELL A sentence it wrote in my home language: Answers vary.